

Finite-Time Feedback-Control Efficiency with Kinetic Compliance

Self-contained derivation plus exact state-local action-to-population information

21 August 2026

Abstract

We derive the finite-time thermodynamic efficiency of a feedback controller acting on a binary population, using the notation of the paper with susceptibility χ . The derivation is self-contained: it starts from the population state, sensor and policy, defines a finite-bias finite-compliance microscopic kernel, derives the closed susceptibility χ , introduces an opposing load, constructs forward and reverse feedback path measures, and obtains the KL second-law identity and the bounded efficiency. We then add a separate operational section on the exact state-local action-to-population information $T_\pi(n) = I(U; m | n)$, which quantifies how visible the controller action is in the next population state at each current state n . The report includes exact plots of this landscape and parameter sweeps over h_c , q_c , β , and θ .

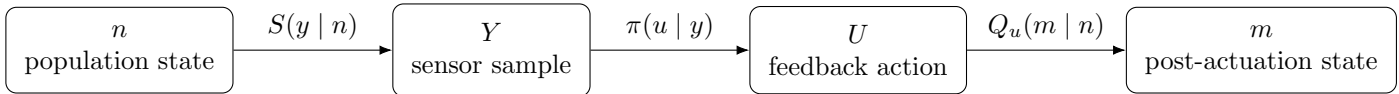
1 Problem and one-round architecture

Consider a population of N binary agents. Fix a controller target Z and let

$$n \in \{0, 1, \dots, N\}, \quad x = \frac{n}{N} \quad (1)$$

be the number and fraction of agents currently supporting Z .

A single feedback cycle has the causal order



The variables and parameters used below are:

N	population size
$n, x = n/N$	target count and target fraction before the cycle
q_c	number of agents sampled by the controller
Y	number of target supporters in the sensor sample
$U \in \{0, 1\}$	feedback action: 0 = no advocacy, 1 = advocate target Z
θ, β	feedback threshold and policy gain
b	number of controlled microscopic opportunities when $U = 1$
h_c	controller input affinity, i.e. directional strength of controlled influence
h_e	opposing load affinity
$\gamma \in [0, 1]$	kinetic compliance: scale of the probability that a controlled opportunity produces a revision
$F = h_c - h_e$	net affinity acting on the controlled revision channel
χ	controller-induced susceptibility of the mean target fraction
J_c	mean net controlled current, measured in target-count changes
$T_\pi(n)$	exact state-local action-to-population information, in bits

All affinities and work-like quantities are dimensionless and all thermodynamic logarithms are natural logarithms.

Scope. The thermodynamic theorem below is for the *feedback-actuation/load layer*. We do not assign a mechanism-resolved thermodynamics to the complete language-model population or to the irreversible social-voter channel. For the behavioral response, however, the ordinary $q = 1$ voter dynamics can be retained because their mean target count is a martingale. This fact is what allows the same exact closed susceptibility to be used for the full advocacy-vs-NoOp mean contrast.

2 Sensing and feedback decision

The controller samples q_c agents without replacement. If Y is the number of target supporters in the sample,

$$Y \mid n \sim \text{Hypergeom}(N, n, q_c), \quad (2)$$

with sensor kernel

$$S(y \mid n) = \frac{\binom{n}{y} \binom{N-n}{q_c-y}}{\binom{N}{q_c}}. \quad (3)$$

After observing $Y = y$, the controller advocates with probability

$$\pi(1 \mid y) = \sigma \left[\beta \left(\theta - \frac{y}{q_c} \right) \right], \quad \sigma(z) = \frac{1}{1 + e^{-z}}. \quad (4)$$

Therefore the advocacy probability conditional on the true population state is the finite sum

$$a_n \equiv \mathbb{P}(U = 1 \mid n) = \sum_y S(y \mid n) \pi(1 \mid y). \quad (5)$$

This is the only place where q_c , θ , and β enter the action probability.

3 Finite-bias, finite-compliance microscopic actuation

It is useful to first write a controlled kernel for a generic directional affinity h . Define

$$p_h = \sigma(h). \quad (6)$$

At one controlled microscopic opportunity, the focal agent is selected uniformly. If it is currently non-target, it revises to the target with probability γp_h ; if it is currently target, it revises away with probability $\gamma(1 - p_h)$. Hence

$$K_{h,\gamma}(n+1 | n) = \gamma \frac{N-n}{N} p_h, \quad (7)$$

$$K_{h,\gamma}(n-1 | n) = \gamma \frac{n}{N} (1 - p_h), \quad (8)$$

$$K_{h,\gamma}(n | n) = 1 - K_{h,\gamma}(n+1 | n) - K_{h,\gamma}(n-1 | n). \quad (9)$$

The two parameters have distinct meanings: h controls the *directional odds*, whereas γ controls the *kinetic probability scale* of revision.

For the behavioral controller we will set $h = h_c$. For the loaded thermodynamic actuation channel we will later set

$$h = F = h_c - h_e. \quad (10)$$

4 Exact susceptibility: derivation of $\chi_{h_c,\gamma}(x)$

This section derives directly the formula

$$\boxed{\chi_{h_c,\gamma}(x) = [\sigma(h_c) - x] \Lambda_{b,\gamma}}, \quad \Lambda_{b,\gamma} = 1 - \left(1 - \frac{\gamma}{N}\right)^b. \quad (11)$$

4.1 One controlled opportunity

Suppose the target fraction immediately before one controlled event is x_j . The count changes by $+1$ with probability

$$\mathbb{P}(\Delta n = +1 | x_j) = \gamma(1 - x_j)p_h, \quad (12)$$

and by -1 with probability

$$\mathbb{P}(\Delta n = -1 | x_j) = \gamma x_j(1 - p_h). \quad (13)$$

Therefore

$$\mathbb{E}[\Delta n | x_j] = \gamma(1 - x_j)p_h - \gamma x_j(1 - p_h) \quad (14)$$

$$= \gamma[p_h - x_j]. \quad (15)$$

Since $x = n/N$,

$$\mathbb{E}[\Delta x | x_j] = \frac{\gamma}{N} [p_h - x_j]. \quad (16)$$

Thus the conditional mean after one controlled opportunity is

$$\mathbb{E}[x_{j+1} | x_j] = x_j + \frac{\gamma}{N} (p_h - x_j) \quad (17)$$

$$= \left(1 - \frac{\gamma}{N}\right) x_j + \frac{\gamma}{N} p_h. \quad (18)$$

Define

$$\alpha_\gamma = 1 - \frac{\gamma}{N}. \quad (19)$$

Then the mean recursion is simply

$$\mathbb{E}[x_{j+1} | x_j] = \alpha_\gamma x_j + (1 - \alpha_\gamma) p_h. \quad (20)$$

4.2 Iterating exactly b controlled opportunities

Let $m_j = \mathbb{E}[x_j \mid x_0 = x]$. Because the recursion is affine,

$$m_{j+1} - p_h = \alpha_\gamma(m_j - p_h). \quad (21)$$

Iterating gives

$$m_b - p_h = \alpha_\gamma^b(x - p_h), \quad (22)$$

so

$$m_b = p_h + (x - p_h)\alpha_\gamma^b \quad (23)$$

$$= x + (p_h - x)(1 - \alpha_\gamma^b). \quad (24)$$

Consequently the mean displacement created by the b controlled opportunities is

$$\mathbb{E}[x_b - x \mid x] = (p_h - x) \left[1 - \left(1 - \frac{\gamma}{N} \right)^b \right]. \quad (25)$$

For the ordinary $q = 1$ voter update,

$$\mathbb{E}[x_{j+1} \mid x_j, \text{ordinary}] = x_j. \quad (26)$$

Hence ordinary updates inserted between the b controlled opportunities do not alter the mean recursion. The full advocacy-vs-NoOp susceptibility is therefore exactly

$$\boxed{\chi_{h_c, \gamma}(x) \equiv \mathbb{E}[x' \mid U = 1, x] - \mathbb{E}[x' \mid U = 0, x] = [\sigma(h_c) - x]\Lambda_{b, \gamma}.} \quad (27)$$

This shows directly what the kinetic term does: γ changes the amplitude through $\Lambda_{b, \gamma}$, while the zero-response point

$$x^* = \sigma(h_c) \quad (28)$$

is determined only by the directional affinity.

5 Introduce the opposing load and obtain the controlled current

Thermodynamic efficiency requires an output against an opposing force. Introduce a dimensionless load affinity $h_e \geq 0$ and define the net controlled affinity

$$F = h_c - h_e, \quad p_F = \sigma(F). \quad (29)$$

The active microscopic kernel is therefore

$$K_{F, \gamma}(n+1 \mid n) = \gamma \frac{N-n}{N} p_F, \quad (30)$$

$$K_{F, \gamma}(n-1 \mid n) = \gamma \frac{n}{N} (1 - p_F). \quad (31)$$

The same calculation as above gives the loaded actuation susceptibility

$$\chi_{F, \gamma}(x) = [p_F - x]\Lambda_{b, \gamma}. \quad (32)$$

Because a change of one in n is one unit of directed count current, the conditional mean current when the controller acts at state n is

$$\mathbb{E}[j_c \mid n, U = 1] = N \chi_{F, \gamma}(n/N) = N \Lambda_{b, \gamma} \left(p_F - \frac{n}{N} \right). \quad (33)$$

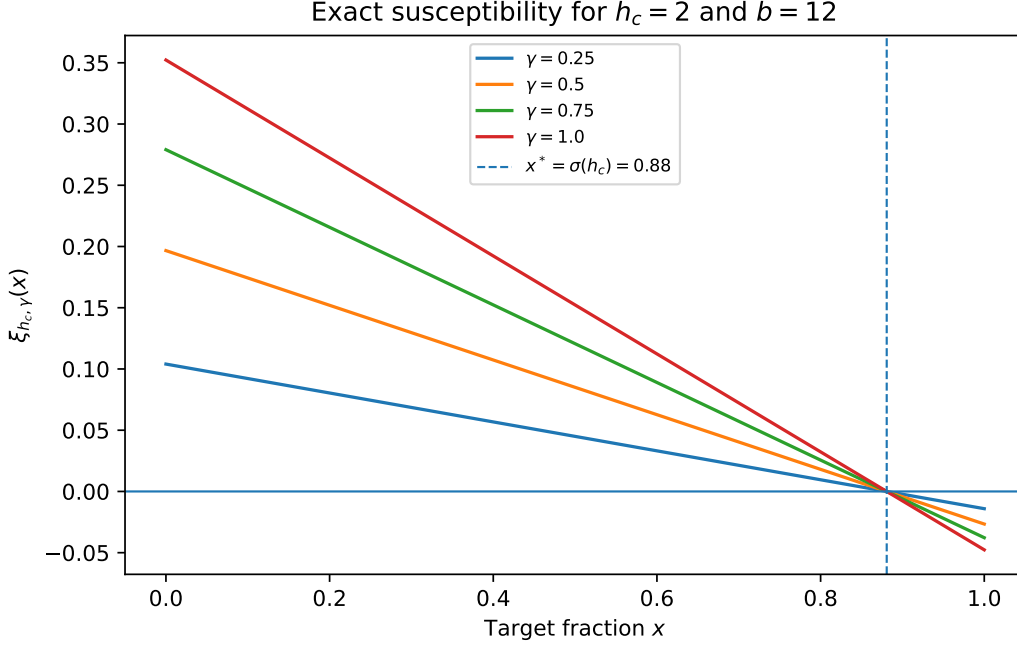


Figure 1: Exact susceptibility for $h_c = 2$ and $b = 12$. Finite kinetic compliance changes the response magnitude without moving the set point $x^* = \sigma(h_c)$.

Under feedback, advocacy occurs with probability a_n , and the pre-measurement population state has distribution $p(n)$. Averaging gives the exact current

$$J_c = N \sum_{n=0}^N p(n) a_n \chi_{F, \gamma}(n/N) = N \Lambda_{b, \gamma} \sum_{n=0}^N p(n) a_n \left[\sigma(h_c - h_e) - \frac{n}{N} \right]. \quad (34)$$

This is the first closed ingredient of the thermodynamic efficiency.

We associate dimensionless generalized input and output work with the same current:

$$W_{\text{ctrl}} = h_c J_c, \quad W_{\text{out}} = h_e J_c. \quad (35)$$

For useful target-directed operation we require $J_c > 0$ and $h_e > 0$.

6 Transition-ratio identity needed for thermodynamics

The key algebraic property is that γ affects kinetics but not the forward/reverse bias. For neighboring count states,

$$\frac{K_{F, \gamma}(n+1 | n)}{K_{F, \gamma}(n | n+1)} = \frac{\gamma \frac{N-n}{N} p_F}{\gamma \frac{n+1}{N} (1-p_F)} \quad (36)$$

$$= \frac{N-n}{n+1} \frac{p_F}{1-p_F} \quad (37)$$

$$= \frac{N-n}{n+1} e^F. \quad (38)$$

The kinetic factor γ cancels exactly. Define the combinatorial macrostate entropy

$$S_{\text{mix}}(n) = \ln \binom{N}{n}. \quad (39)$$

Since

$$S_{\text{mix}}(n+1) - S_{\text{mix}}(n) = \ln \frac{N-n}{n+1}, \quad (40)$$

we obtain

$$\ln \frac{K_{F,\gamma}(n+1 | n)}{K_{F,\gamma}(n | n+1)} = S_{\text{mix}}(n+1) - S_{\text{mix}}(n) + F. \quad (41)$$

For later use define the positive weight

$$w_F(n) = \binom{N}{n} e^{Fn}. \quad (42)$$

The previous identity is equivalent to

$$w_F(n) K_{F,\gamma}(m | n) = w_F(m) K_{F,\gamma}(n | m) \quad (43)$$

for every supported transition. This algebraic reversibility relation is the only property needed in the path derivation.

If the active layer contains exactly b controlled opportunities, define

$$Q_1 = K_{F,\gamma}^b, \quad Q_0 = I. \quad (44)$$

Because the reversibility identity is preserved under matrix powers,

$$w_F(n) Q_1(m | n) = w_F(m) Q_1(n | m), \quad (45)$$

and therefore

$$\ln \frac{Q_1(m | n)}{Q_1(n | m)} = S_{\text{mix}}(m) - S_{\text{mix}}(n) + F(m - n). \quad (46)$$

This endpoint identity is what makes the finite-time path derivation possible without resolving every microscopic controlled jump.

7 Forward and reverse feedback path measures

Let $p(n)$ be the distribution of the population count before sensing. A forward feedback realization is

$$n \longrightarrow y \longrightarrow u \longrightarrow m, \quad (47)$$

with joint probability

$$P_F(n, y, u, m) = p(n) S(y | n) \pi(u | y) Q_u(m | n). \quad (48)$$

The final population marginal is

$$p'(m) = \sum_{n,y,u} P_F(n, y, u, m). \quad (49)$$

The marginal distribution of the sensor record is

$$p_Y(y) = \sum_n p(n) S(y | n). \quad (50)$$

Choose the reverse reference experiment as

$$P_R(m, y, u, n) = p'(m) p_Y(y) \pi(u | y) Q_u(n | m). \quad (51)$$

Thus the reverse experiment starts from the forward final marginal, draws the stored protocol label y independently from its forward marginal, applies the same feedback rule $\pi(u | y)$, and then uses the reversed controlled kernel. This particular construction is designed so that the feedback-policy factor cancels in the path ratio.

8 Path ratio and the finite-time feedback second law

Define the log path-weight ratio

$$\Sigma^*(n, y, u, m) = \ln \frac{P_F(n, y, u, m)}{P_R(m, y, u, n)}. \quad (52)$$

Substituting the two path measures gives

$$\Sigma^* = \ln \frac{p(n)}{p'(m)} + \ln \frac{S(y | n)}{p_Y(y)} + \ln \frac{Q_u(m | n)}{Q_u(n | m)}. \quad (53)$$

The factor $\pi(u | y)$ cancels exactly.

Define the pointwise sensing information

$$i(n; y) = \ln \frac{S(y | n)}{p_Y(y)}, \quad (54)$$

and the stochastic system entropy

$$s_{\text{sys}}(n; p) = -\ln p(n) + S_{\text{mix}}(n). \quad (55)$$

For supported paths define the controlled current increment

$$j_c = u(m - n). \quad (56)$$

For $u = 0$, $Q_0 = I$, hence $m = n$ and $j_c = 0$. For $u = 1$, the endpoint transition-ratio identity above applies. Therefore

$$\ln \frac{p(n)}{p'(m)} + S_{\text{mix}}(m) - S_{\text{mix}}(n) = s_{\text{sys}}(m; p') - s_{\text{sys}}(n; p), \quad (57)$$

and the path ratio becomes

$$\boxed{\Sigma^* = \Delta s_{\text{sys}} + F j_c + i(n; y)}. \quad (58)$$

This is the central pathwise identity.

Now average over the forward path ensemble. By construction,

$$\langle \Sigma^* \rangle_F = D_{\text{KL}}(P_F \| P_R) \geq 0. \quad (59)$$

The three average terms are

$$\Delta S_{\text{sys}} = S_{\text{sys}}[p'] - S_{\text{sys}}[p], \quad (60)$$

$$J_c = \mathbb{E}_F[j_c], \quad (61)$$

$$I_{\text{sens}} = \mathbb{E}_F[i(n; Y)] = I(n; Y). \quad (62)$$

The system-entropy functional is

$$\boxed{S_{\text{sys}}[p] = -\sum_n p(n) \ln p(n) + \sum_n p(n) \ln \binom{N}{n}}. \quad (63)$$

The sensing information has the exact finite-sum form

$$\boxed{I_{\text{sens}} = \sum_{n, y} p(n) S(y | n) \ln \frac{S(y | n)}{p_Y(y)}}. \quad (64)$$

Hence

$$\boxed{\Delta S_{\text{sys}} + (h_c - h_e) J_c + I(n; Y) = D_{\text{KL}}(P_F \| P_R) \geq 0}. \quad (65)$$

This is the finite-time feedback second law for the reversible actuation/load layer.

9 Closed finite-time thermodynamic efficiency

Rearrange the KL identity:

$$\Delta S_{\text{sys}} + h_c J_c - h_e J_c + I(n; Y) = D_{\text{KL}}(P_F \| P_R), \quad (66)$$

$$h_e J_c = h_c J_c + I(n; Y) + \Delta S_{\text{sys}} - D_{\text{KL}}(P_F \| P_R). \quad (67)$$

Because the KL divergence is non-negative,

$$\boxed{h_e J_c \leq h_c J_c + I(n; Y) + \Delta S_{\text{sys}}.} \quad (68)$$

For $h_e > 0$ and $J_c > 0$, define useful output divided by total finite-time resource accounting:

$$\boxed{\eta_{\text{th}} = \frac{h_e J_c}{h_c J_c + I(n; Y) + \Delta S_{\text{sys}}} \leq 1.} \quad (69)$$

Using the exact KL equality, the same efficiency can be written as

$$\boxed{\eta_{\text{th}} = \frac{h_e J_c}{h_e J_c + D_{\text{KL}}(P_F \| P_R)}. \quad (70)$$

The bound is now immediate: the denominator is useful output plus a non-negative irreversibility term.

9.1 Everything in the efficiency is computable by finite sums

The current is already available in closed form:

$$J_c = N \Lambda_{b,\gamma} \sum_n p(n) a_n \left[p_F - \frac{n}{N} \right], \quad p_F = \sigma(h_c - h_e). \quad (71)$$

The final state distribution required for ΔS_{sys} is

$$\boxed{p'(m) = \sum_n p(n) [(1 - a_n) \delta_{mn} + a_n Q_1(m | n)], \quad Q_1 = K_{F,\gamma}^b.} \quad (72)$$

Therefore the complete exact finite- N efficiency is

$$\boxed{\eta_{\text{th}} = \frac{h_e N \Lambda_{b,\gamma} \sum_n p(n) a_n [p_F - n/N]}{h_c N \Lambda_{b,\gamma} \sum_n p(n) a_n [p_F - n/N] + I(n; Y) + S_{\text{sys}}[p'] - S_{\text{sys}}[p]}. \quad (73)$$

There is no Monte Carlo approximation in this expression. For given $N, b, q_c, \theta, \beta, h_c, h_e, \gamma$ and initial state distribution $p(n)$, every term is obtained by finite sums and the finite matrix power $K_{F,\gamma}^b$.

10 A direct computation recipe

For an empirical or theoretical ensemble with pre-sensing state distribution $p(n)$, the derivation above reduces to the following sequence:

1. Compute the sensor kernel $S(y | n)$ and policy $\pi(1 | y)$.
2. Compute $a_n = \sum_y S(y | n) \pi(1 | y)$ and $p_Y(y) = \sum_n p(n) S(y | n)$.

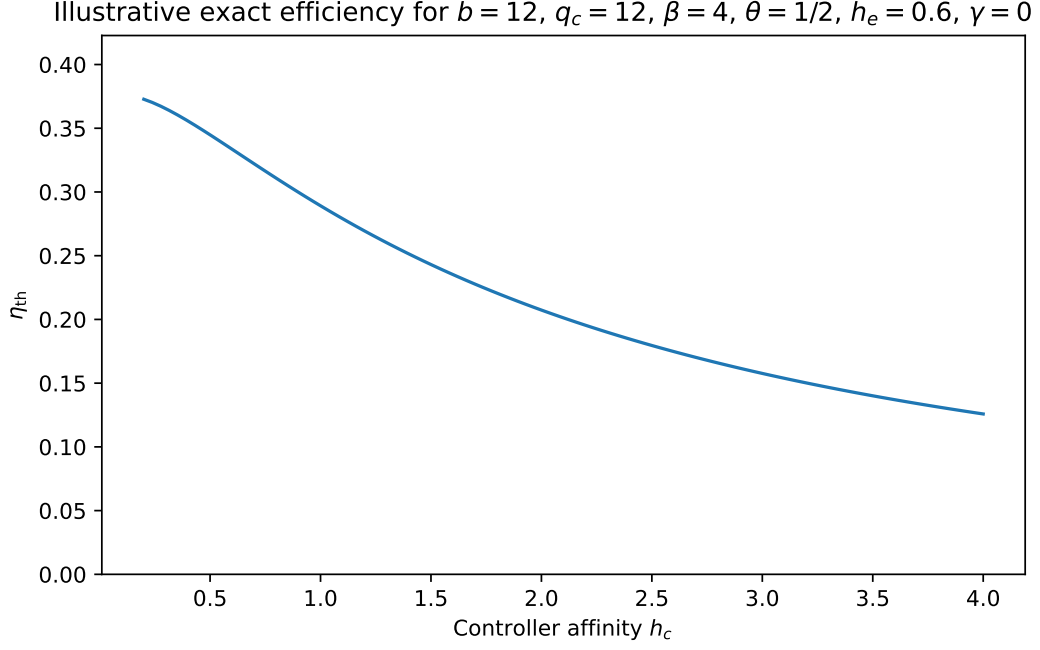


Figure 2: Illustrative exact finite-time efficiency for one chosen initial ensemble $p(n) = \text{Binomial}(N, 0.25)$. The horizontal line marks the theorem bound $\eta_{th} = 1$. The plot is only illustrative; the derivation itself does not require fixing a particular $p(n)$.

3. Set $F = h_c - h_e$, $p_F = \sigma(F)$, and $\Lambda_{b,\gamma} = 1 - (1 - \gamma/N)^b$.
4. Build $K_{F,\gamma}$ and $Q_1 = K_{F,\gamma}^b$.
5. Compute the mean current

$$J_c = N\Lambda_{b,\gamma} \sum_n p(n) a_n (p_F - n/N).$$

6. Compute

$$I(n; Y) = \sum_{n,y} p(n) S(y | n) \ln \frac{S(y | n)}{p_Y(y)}.$$

7. Compute

$$p'(m) = \sum_n p(n) [(1 - a_n) \delta_{mn} + a_n Q_1(m | n)].$$

8. Compute $\Delta S_{\text{sys}} = S_{\text{sys}}[p'] - S_{\text{sys}}[p]$ using

$$S_{\text{sys}}[p] = - \sum_n p(n) \ln p(n) + \sum_n p(n) \ln \binom{N}{n}.$$

9. Finally evaluate

$$\eta_{th} = \frac{h_e J_c}{h_c J_c + I(n; Y) + \Delta S_{\text{sys}}}.$$

The KL identity guarantees the bound whenever $J_c > 0$ and $h_e > 0$.

11 Role of kinetic compliance

The kinetic parameter γ enters the efficiency through the transition kinetics, not through the thermodynamic affinity. At the mean-response level it appears through

$$\Lambda_{b,\gamma} = 1 - \left(1 - \frac{\gamma}{N}\right)^b, \quad (74)$$

which controls the current magnitude. At the path level it changes $Q_1 = K_{F,\gamma}^b$ and therefore the final distribution p' and the KL divergence. However, because the same constant γ multiplies the forward and reverse microscopic revision rates, it cancels from their ratio. No additional $\ln \gamma$ term appears in the affinity or in the feedback second law.

At $\gamma = 0$, $K_{F,0} = I$ and $J_c = 0$, so there is no useful motor operation. For $0 < \gamma \leq 1$, the complete derivation above applies. In the limit $h_c \rightarrow \infty$, directionality becomes perfect, but actuation remains kinetically incomplete unless $\gamma = 1$.

12 Exact state-local action-to-population information

We now add the quantity illustrated in your reference image, but written for the present finite-compliance theory. This object is *not* part of the thermodynamic efficiency formula. Instead, it is a complementary operational diagnostic that tells us how visible the controller action is in the next population state.

12.1 Definition

For a fixed pre-actuation state n , define the two post-actuation kernels

$$Q_0(m | n) = \delta_{mn}, \quad Q_1(m | n) = K_{h_c,\gamma}^b(m | n). \quad (75)$$

The state-dependent action probability is $a_n = \mathbb{P}(U = 1 | n)$. The action-marginalized next-state kernel is therefore

$$Q_\pi(m | n) = (1 - a_n)Q_0(m | n) + a_nQ_1(m | n). \quad (76)$$

The exact state-local action-to-population information is the conditional mutual information at that state,

$$T_\pi(n) \equiv I(U; m | n) = \sum_{u \in \{0,1\}} \mathbb{P}(u | n) \sum_m Q_u(m | n) \log_2 \frac{Q_u(m | n)}{Q_\pi(m | n)}. \quad (77)$$

Hence $T_\pi(n)$ is measured in *bits*. It becomes large only when two conditions are met simultaneously:

1. the action is genuinely variable, so that a_n is not too close to 0 or 1;
2. the two kernels $Q_0(\cdot | n)$ and $Q_1(\cdot | n)$ are well separated.

This is exactly why the heatmaps show a ridge rather than a monotone pattern.

If one also averages over a pre-actuation ensemble $p(n)$, the full exact next-state action information is

$$I(U; m | n_{\text{pre}}) = \sum_n p(n) T_\pi(n). \quad (78)$$

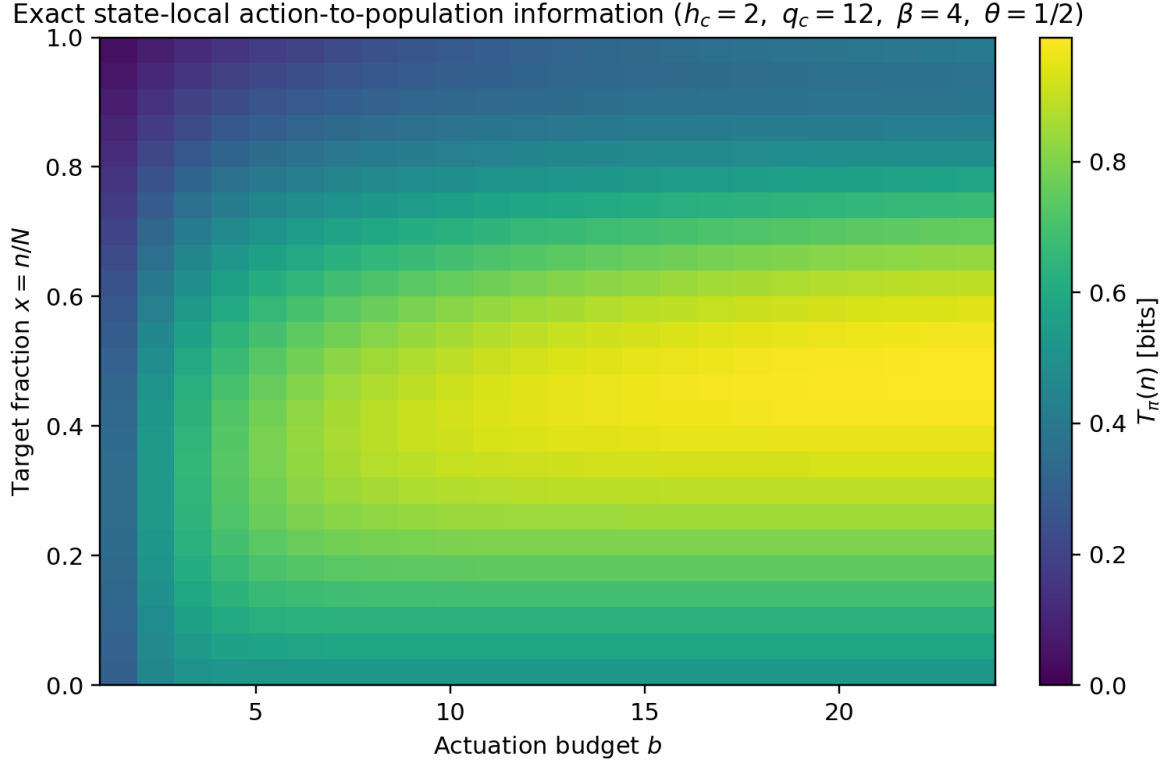


Figure 3: Exact state-local action-to-population information for $N = 24$, $h_c = 2$, $q_c = 12$, $\beta = 4$, $\theta = 1/2$, and $\gamma = 1$. The information increases with b because the advocacy and NoOp kernels separate more strongly, but it stays small where the action is nearly deterministic.

12.2 Baseline landscape

Figure 3 shows the exact local landscape as a function of the current target fraction $x = n/N$ and the actuation budget b , for the baseline values $N = 24$, $h_c = 2$, $q_c = 12$, $\beta = 4$, $\theta = 1/2$, and $\gamma = 1$. The baseline heatmap is the direct analogue of your earlier picture, but now computed from the present finite-compliance kernel formulation. The main qualitative message is unchanged: finite bias preserves a nontrivial state-local information landscape, and the ridge reflects both action variability and kernel separation.

12.3 Parameter sweeps

Figures 4–7 show how the same exact quantity changes when we vary h_c , q_c , β , and θ one by one while keeping the other parameters at their baseline values.

12.4 How to read the parameter dependence

The exact sweeps support the following interpretation.

1. **Increasing h_c .** This strengthens directional advocacy, so the separation between Q_1 and Q_0 increases. As a result, the local information generally rises.
2. **Changing q_c .** This changes how informative the sample is about n . A larger q_c sharpens the map $n \mapsto a_n$, which can either increase or decrease $T_\pi(n)$ depending on whether it creates a broad band of partial action uncertainty or a nearly deterministic policy.

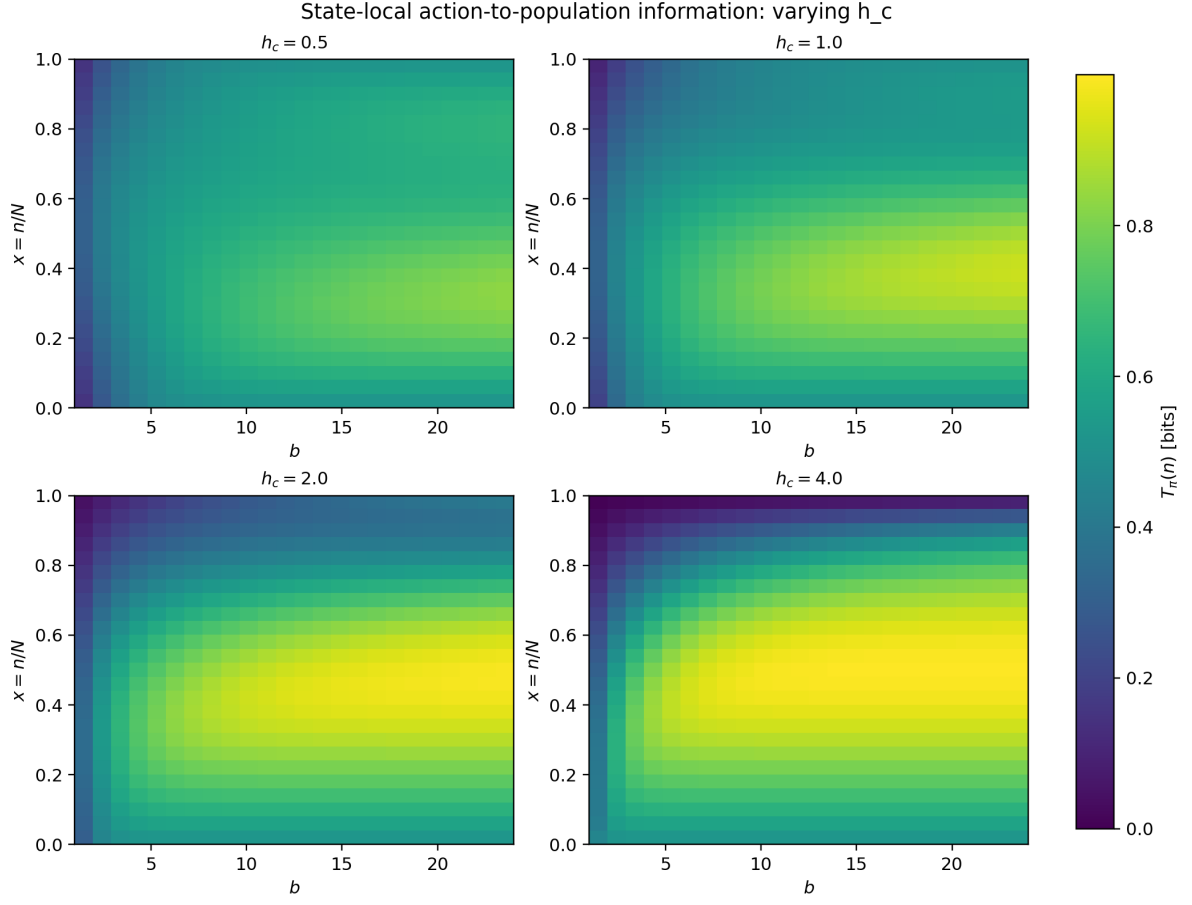


Figure 4: Exact state-local action-to-population information while varying h_c . Larger h_c separates the advocacy kernel more strongly from NoOp and therefore raises $T_\pi(n)$, but the enhancement is concentrated in states where the policy still has nontrivial action entropy.

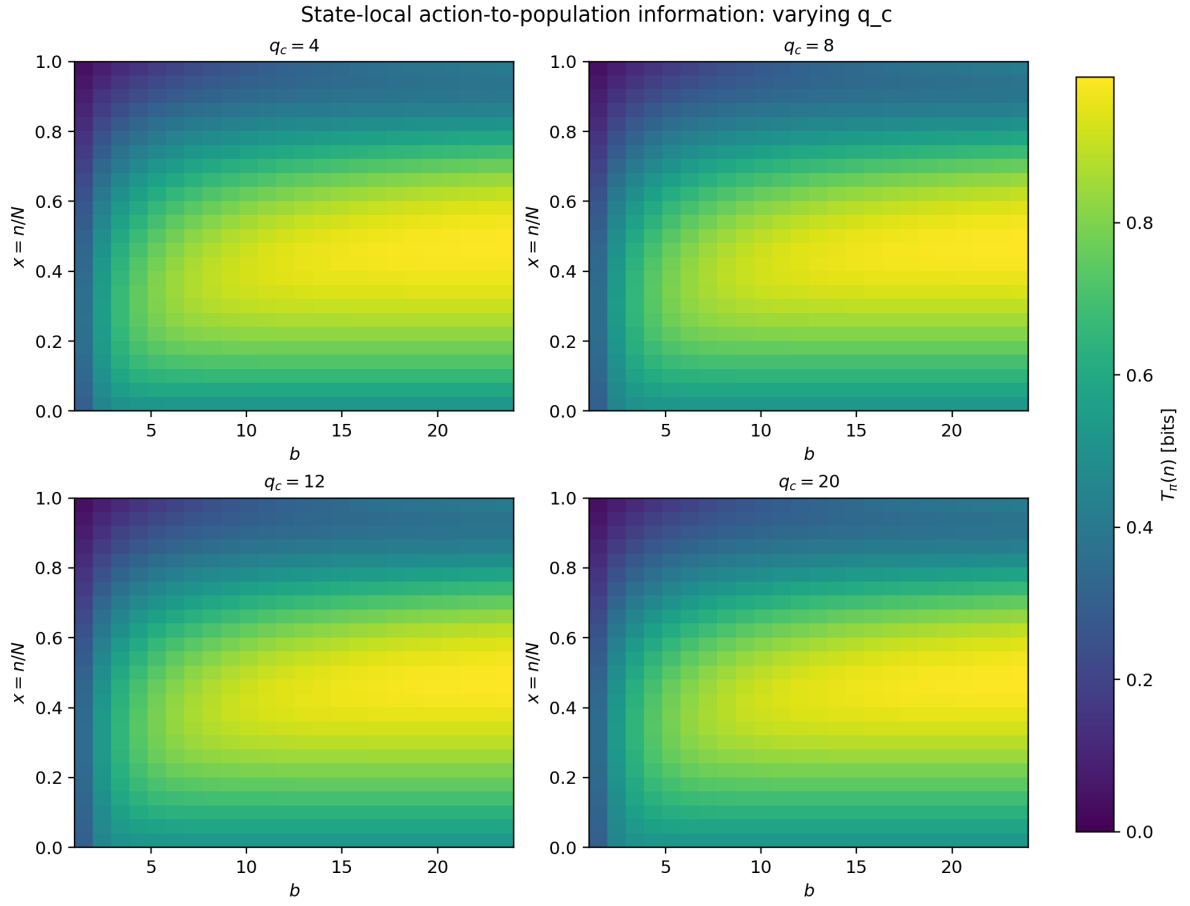


Figure 5: Exact state-local action-to-population information while varying q_c . Changing q_c changes how sharply the controller can resolve the current target fraction before acting, which in turn changes the state dependence of a_n and therefore the ridge structure of $T_\pi(n)$.

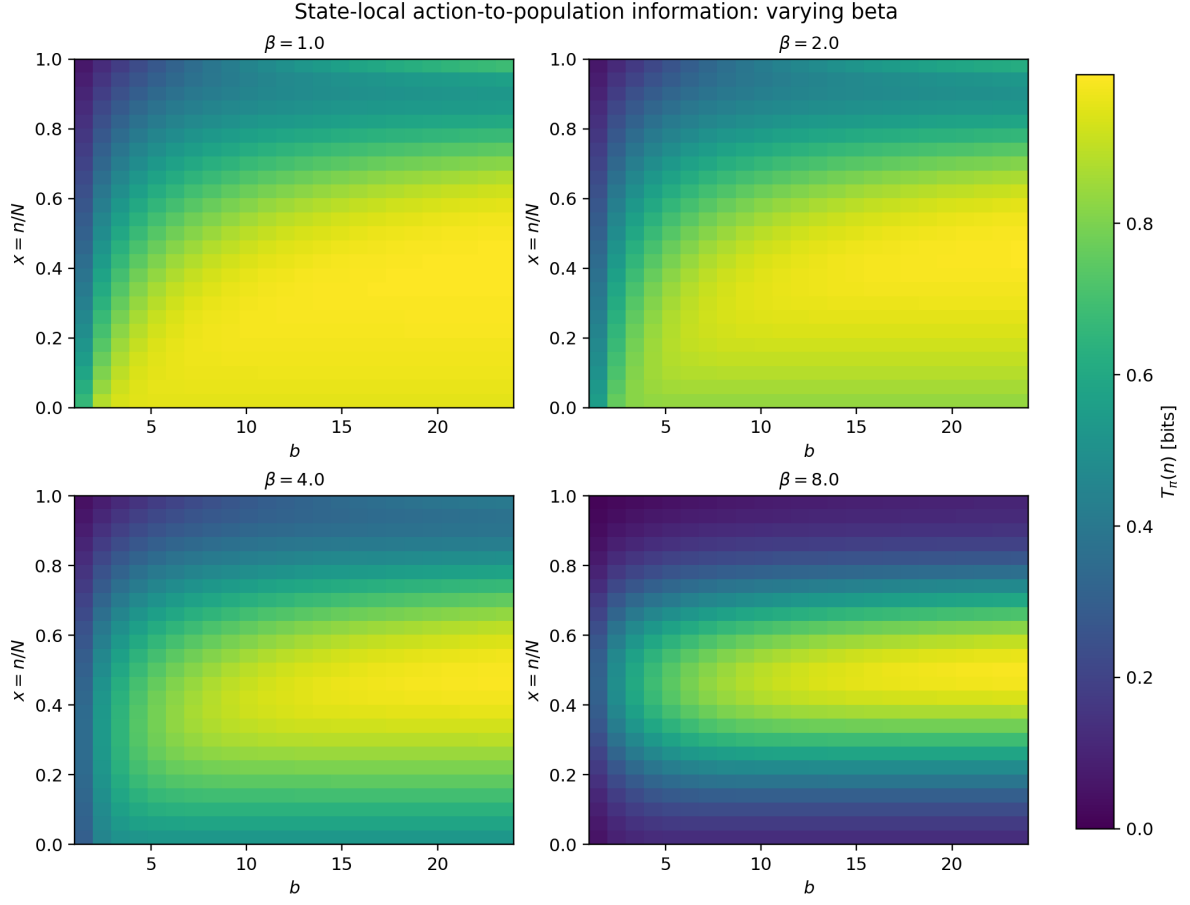


Figure 6: Exact state-local action-to-population information while varying β . Larger β makes the policy more switch-like. This sharpens the informative band at intermediate states but also suppresses information where the action saturates to near-certain advocacy or near-certain NoOp.

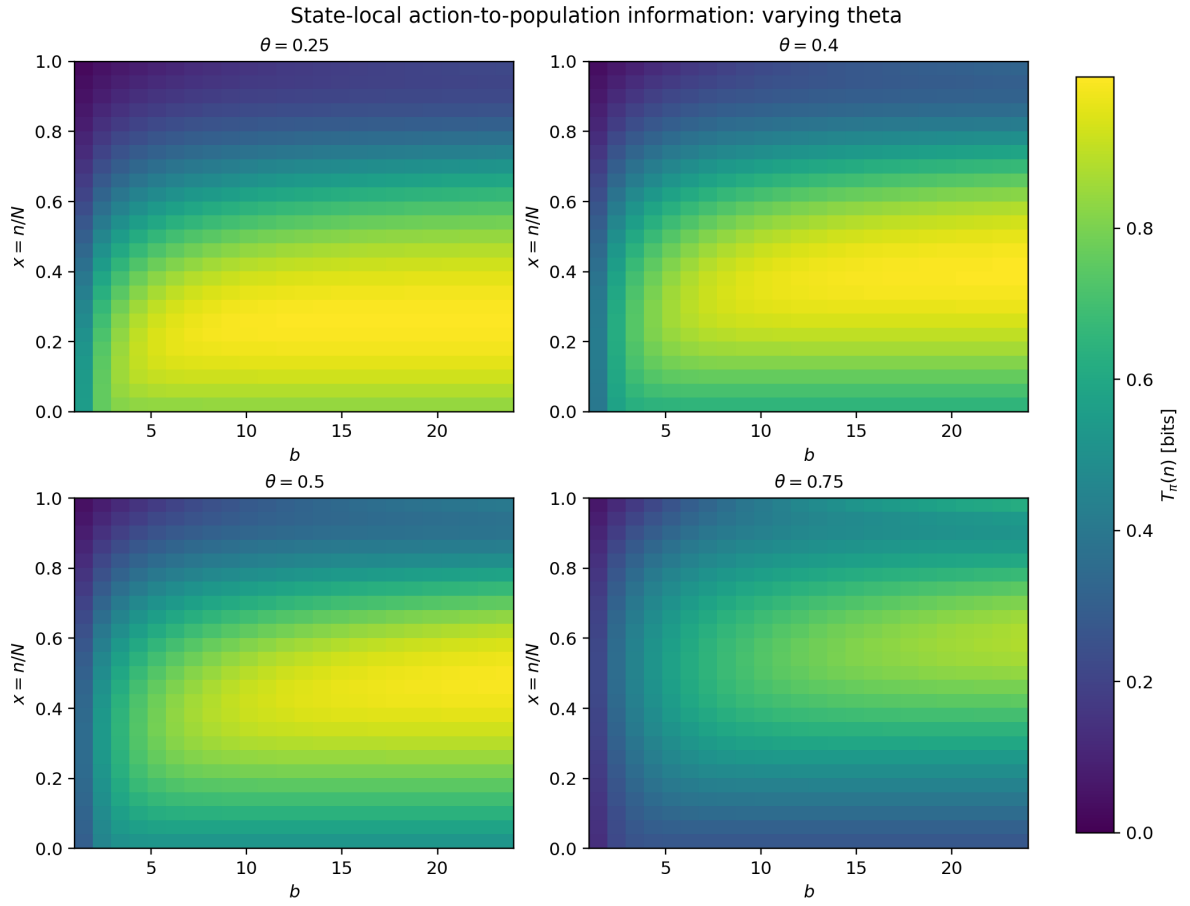


Figure 7: Exact state-local action-to-population information while varying θ . The threshold shifts the region in state space where the controller is undecided, which correspondingly shifts the high-information ridge.

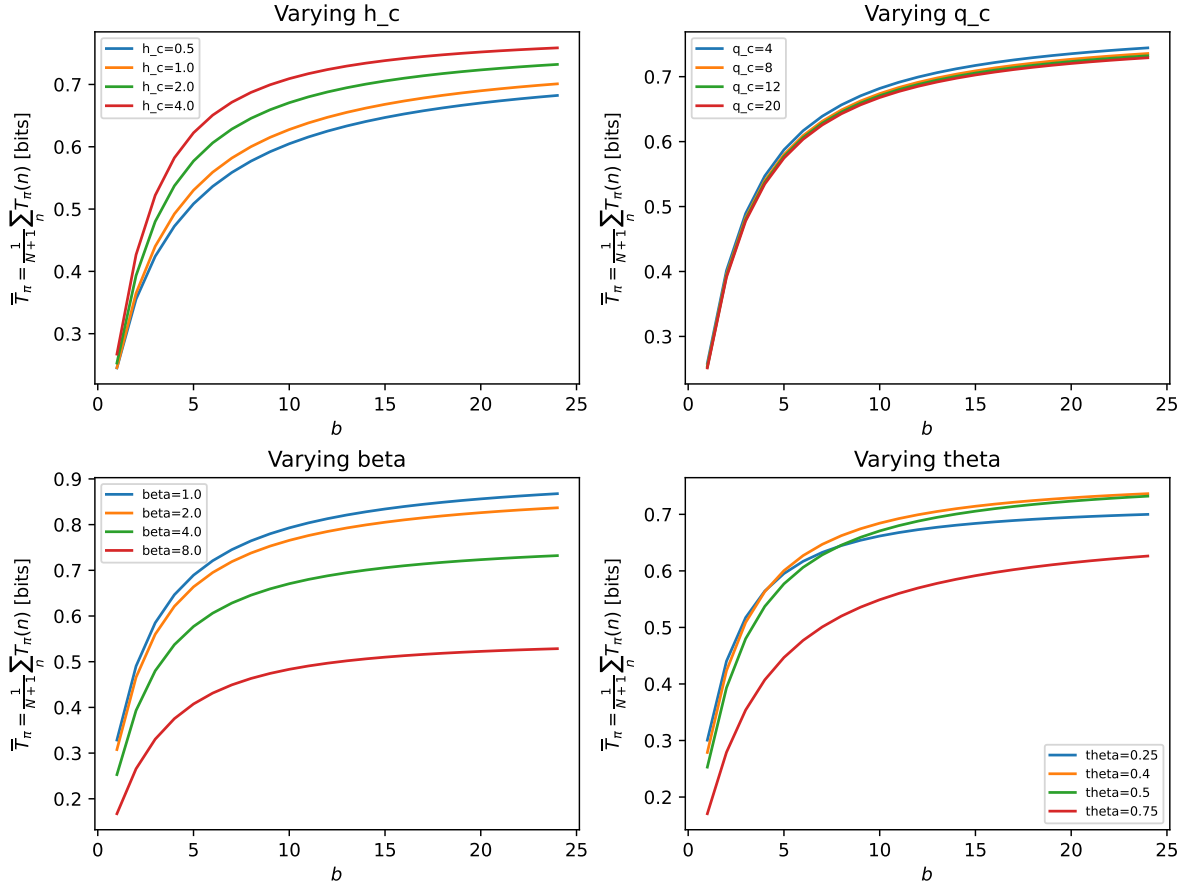


Figure 8: State-averaged local information $\bar{T}_\pi = (N + 1)^{-1} \sum_n T_\pi(n)$ as a compact summary of the landscape. This figure is useful when one wants a single curve versus b for each parameter family, while the heatmaps remain the main object for understanding the state dependence.

3. **Increasing β .** A steeper policy makes the action more state-selective. Around the switching region this can increase $T_\pi(n)$, but away from that region it can drive the action entropy to zero.
4. **Changing θ .** The threshold shifts the switching region along the x -axis, so the high-information ridge moves accordingly.

In short, h_c controls *kernel separation*, whereas q_c , β , and θ control *action variability through sensing and policy*. The local information is large only when both ingredients are present.

13 Connection to microscopic LLM transitions

If controlled microscopic transitions are summarized by

$$p_+ = \mathbb{P}(\text{non-}Z \rightarrow Z \mid \text{controlled}), \quad (79)$$

$$p_- = \mathbb{P}(Z \rightarrow \text{non-}Z \mid \text{controlled}), \quad (80)$$

then the finite-compliance kernel implies

$$p_+ = \gamma\sigma(F), \quad p_- = \gamma\sigma(-F). \quad (81)$$

Thus

$$\boxed{\gamma_{\text{eff}} = p_+ + p_-, \quad F_{\text{eff}} = \ln \frac{p_+}{p_-}.} \quad (82)$$

For the observed transition frequencies $p_+ = 208/572 \approx 0.364$ and $p_- = 4/508 \approx 0.0079$, this gives approximately

$$\gamma_{\text{eff}} \approx 0.372, \quad F_{\text{eff}} \approx 3.83. \quad (83)$$

These quantities calibrate the net actuation channel. They do not by themselves identify h_c and h_e separately because the controlled transition odds determine their difference $F = h_c - h_e$.

14 Scope of the derivation

The derivation requires only: the finite-state population coordinate, the sensor and policy, the finite-compliance controlled kernel, the load decomposition $F = h_c - h_e$, the transition-ratio identity, and the forward/reverse path construction. No additional dynamical construction is required beyond these objects.

The exact action-to-population information $T_\pi(n)$ is also separate from η_{th} . It is a downstream diagnostic of how visibly the action changes the next-state distribution. The information resource in the feedback second law is instead the information acquired by the sensor, $I(n; Y)$.

References

- [1] J. M. Horowitz and H. Sandberg, “Second-law-like inequalities with information and their interpretations,” *New Journal of Physics*, 16, 125007 (2014).
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